LOT-HW-01 — Plan & Roadmap (v0.3)

1. PRODUCT

- **Form:** flat silver square, 40 × 40 mm, 2-part 316L stainless body. Target 5 mm; see R1, recommended 5.4 mm max.
- **Side A:** polished mirror stainless, no openings.
- **Side B:** screen, camera (with shutter LED), one button ("Copy").
- **Inside:** ESP32-S3 (Wi-Fi/BLE), 1.54″ e-paper or 0.96″ OLED, BME280/BME688 weather sensor, IMU, light sensor, LiPo, Qi receiver, haptic.
- **Job:** receive AI-written pager notices from LOT; Copy → Log tab; weather → site; charge wirelessly.

2. YOUR 19 POINTS → WHERE THEY ARE HANDLED

#	Requirement	Answer	Status
1	PCBWay	PCB + PCBA + CNC steel quotes, doc 06	Plan
2	Pager notice from AI site	`queueDeviceNotice()` + `/api/device/inbox`	**Built**
3	2-part stainless body	Back shell + front bezel, 316L, doc 06	Plan
4	4×4 cm × 5 mm	40×40 OK; 5.0 mm only without camera/Qi → R1	Decision
5	Camera	OV2640/GC2145, must be ≤ 3.5 mm tall for slim build	Sourcing
6	LOT API connector	doc 03, `device-api.ts`	**Built**
7	PDF manuals	`pdf/` regenerated each session	**Built**
8	Compress each session	`sessions/` one-page reports	Ongoing
9	Firmware docs	doc 04	v0.1
10	Software ↔ firmware	simulator/CLI, doc 05	**Built**
11	Separate documents	8 docs + reports	Done
12	Charger	Off-the-shelf certified 5 W Qi pad	Plan
13	100-unit run	doc 06	Plan
14	Weather sensor	BME280 (EVT) → BME688 (DVT)	**Built in API + FW**
15	AI-grade off-the-shelf sensors	BME688 (BSEC AI gas), LSM6DSO, VEML7700	BOM
16	Copy → Log tab	`device_copy` event, `PAGER:` row	**Built**
17	One polished side	Side A	Plan
18	Camera + screen + button side	Side B	Plan
19	Wireless charger	Qi RX through a glass window (R2)	Decision

3. SYSTEM FLOW

```
LOT site AI / scheduled job ── queueDeviceNotice(userId,"Coffee time!") ──►
logs(event=device_notice, pending)
                                                                          │
Pager ── GET /api/device/inbox (every 60 s) ◄─────────────────────────────┘
  ├─ screen + buzz ── POST /api/device/ack ──► notice marked delivered
  ├─ button "Copy"  ── POST /api/device/event {type:copy} ──► logs(event=device_copy) ──► Log tab
"PAGER:"
  └─ every 15 min   ── POST /api/device/event {type:sensor} ──► logs(event=device_sensor, hidden)
```

4. MECHANICAL STACK (PLANNING VALUES, NOT DATASHEET-CHECKED)

Layer	Full	Slim (rec.)
Steel back (A)	0.8	0.6
Battery	3.0 (≈250 mAh)	2.0 (≈130 mAh)
PCB + parts	1.6	1.2
Display	1.0	0.9
Front glass/bezel (B)	1.0	0.7
Total	**7.4**	**5.4**

Qi coil sits beside the battery under the front glass, so it adds no height. The camera must be ≤ 3.5 mm or sit in a lens pocket.

5. ROADMAP (ANALYSIS)

Phase	Weeks	Window	Exit	Cash (est.)
P0 Decisions	1–2	Nov 11–24	R1/R2/R3 signed	$0
P1 Breadboard	3	Nov 18–Dec 9	Devkit + e-paper + BME280 runs firmware v0.1 against site	$150–300
P2 Site API	—	**done S3**	Deploy migration + routes; pair real token	dev time
P3 EVT (5 boards)	8	Dec 9–Feb 3	KiCad schematic/PCB, PCBWay PCBA, 1 CNC steel shell	$1.5–2.5k
P4 DVT (20–30)	8	Feb 3–Mar 31	Final body, Qi through glass ≥ 60 % eff., 72 h soak, drop test	$5–8k
P5 Compliance pre-scan	4	Mar 31–Apr 28	FCC/CE pre-scan, UN38.3 path	$3–10k
P6 PVT 100 units	8	Apr 28–Jun 23	100 pass jig, signed OTA, PDF manuals in box	$5–7k

- **Critical path:** R1/R2 sign-off → steel CNC + polish spins (3–5 weeks each) → Qi through glass → compliance.
- **Realistic:** first 100 units **Jun–Aug 2027**. Expect a 30–40 % slip at DVT.
- **Budget:** about $15–30k all-in (about $150–300 per unit amortised); marginal cost about $63 per unit (BOM sum at qty 100, doc 02).
- **Change since S2:** P2 (site API) is done, about 3 weeks early. The calendar is unchanged because P0 is still open.

6. RISKS

R1 thickness · R2 steel blocks Qi · R3 steel detunes Wi-Fi · R4 heat in a sealed body (charge at 0.5C, NTC) · R5 camera privacy (off by default, LED, no upload in v1) · R6 mirror polish scratches · R7 0.6 mm walls may not stay flat (needs PCBWay DFM) · R8 prices and links are estimates, not live quotes.